

Iowa State University

Iowa Environmental Mesonet

[CONTACT US](#) [DISCLAIMER](#) [APPS](#)

Search "🔍"

[NWS Mainpage](#) / [SPC Convective and Fire Weather + WPC Excessive Rainfall Outlooks](#)

The IEM attempts to process and archive the convective and fire weather outlooks issued by the Storm Prediction Center and the excessive rainfall outlooks issued by the Weather Prediction Center. This processing is done by parsing the PTS product via pyIEM decoder. Recently, the SPC website has added direct single outlook downloads in various GIS formats. **Your best data quality will come from scraping their website**, but maybe that approach is not for all users of this data! So here it is, a means to bulk download the IEM's SPC/WPC Outlook database.

 Backend documentation exists for those wishing to script against this service. Due to processing constraints, requests are limited to 10 single outlook type-years at a time.

Related: [NWS Misc Polygons](#) [NWS VTEC Watch Warning Advisory](#) [SPC Convective Discussions](#) [SPC Watch Polygons](#) [WPC Precip Discussions](#)

Time Interval for Outlook *Issuance* (UTC)

	Year	Month	Day	Hour	Minute	Outlook Type:	Outlook Day:
Start:	2020	Jan	1	0 UTC	0	☒ Convective (1987+)	☒ 1 ☐ 2 ☐ 3 ☐ 4
End:	2024	Dec	31	23 UTC	59	☐ Excessive Rainfall (2019+)	☐ 5 ☐ 6 ☐ 7 ☐ 8
						☐ Fire Weather (2006+)	

☐ Geometries as 'Cake Layers' ☒ Geometries as 'Cookie Cutters'

Shapefile DBF schema:

Field 0: Type=C/String, Title='ISSUE', Outlook Beginning Valid Time UTC
Field 1: Type=C/String, Title='EXPIRE', Outlook Ending Valid Time UTC
Field 2: Type=C/String, Title='PRODISS', Timestamp of Product Issuance UTC
Field 3: Type=C/String, Title='TYPE', Either C=Convective, F=Fire Weather
Field 4: Type=N/Double, Title='DAY', Day 1 through 8 for the outlook
Field 5: Type=C/String, Title='THRESHOLD', Threshold Label
Field 6: Type=C/String, Title='CATEGORY', Category of Threshold Label
Field 7: Type=N/Double, Title='CYCLE', The generalized forecast issuance hour

Archive / Usage Notes:

- 11 Jul 2023:** The archive was reprocessed to allow for specification of returned geometries in either "cake layer" or "cookie cutter" form. This gets complex to explain, but hopefully as a simple example: If you are under a "Moderate Risk", the cake layer form would include polygons with thresholds of general thunderstorm, marginal, slight, and enhanced for your location: the cookie cutter form would just have a moderate polygon covering your location.
- 1 Apr 2021:** An assessment was made of the archive completeness and found 99.9% coverage. Some of the holes are due to lost PTS text products yet to be found and others are due to errors within the PTS product causing it to be generally unusable.
- A complication with using this dataset is that all outlooks are returned, the CYCLE attribute attempts to define a canonical outlook for a given issuance cycle. For example, the Day 1 20Z convective outlook may get issued three times for various reasons. The download contains all three, but two of them will have a cycle of '-1' and the other will have a '20'.
- Presently, the service only returns outlooks that had actual geographies associated with the outlook. For example, if there was no Day 5 area in the outlook, no entries are included in the download result.

IOWA STATE UNIVERSITY

College of Ag

Department of Agronomy

Department of Agronomy

716 Farm House Ln

Ames, IA 50011

akrherz@iastate.edu



Facebook



Twitter



Github



YouTube



RSS

Copyright © 2001-2025

Iowa State University
of Science and Technology
All rights reserved.[Non-discrimination Policy](#)[Privacy Policy](#)[Digital Access & Accessibility](#)

